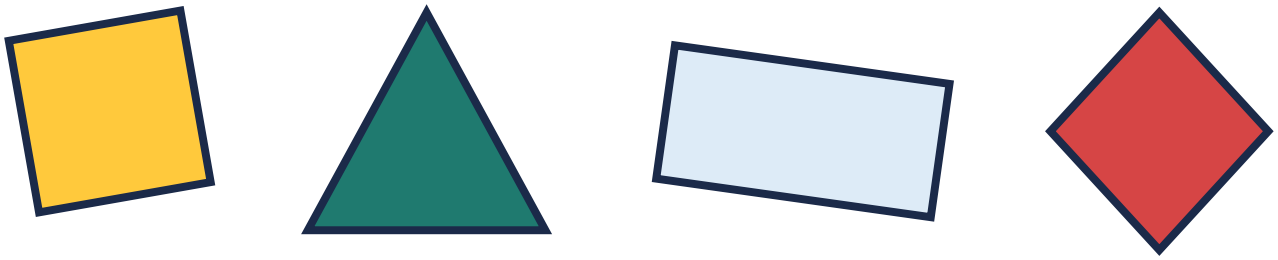
FOR PARENTS • LEVEL 1

Weekly Dose of Thinking Math

One dose a week for five weeks: short sets of questions for every Math Inquirers lesson, a tip for you, and the answers at the back.



Notice it. Test it. Explain why.

Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

00 FOR GROWN-UPS How to help

1. **Ask before you tell.** Start with "What do you notice?" and let your child answer first.
2. **Use real things.** Coins, notes, buttons, spoons, pasta or blocks. Try it with objects before writing.
3. **Give hints in three steps.** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right. The reason matters as much as the answer.
5. **Praise the thinking.** "I like how you checked that" works better than "You are clever".
6. **Keep it short.** Ten minutes and two or three questions is enough. Stop while it is still fun.
7. **Answers are at the back.** Look only after your child has tried.

> **Hard question? Great. Getting stuck is where the thinking happens.**

One dose a week

Each week has two lessons and about 18 questions. Do two or three a day, Monday to Friday, and keep the weekend for the Go further questions if your child enjoys them.

The three levels

Start here is for everyone. **Think harder** and **Go further** stretch the idea, and every child may choose them. The **Investigation** is an open task anyone can try. It is fine to leave any of them for another day.

What your child is doing in class

Ten 90-minute lessons over five weeks. Each lesson starts with a hands-on puzzle, then a short teaching part, then the workbook. These pages practise the same ideas at home.

W1 LESSON 1 • NUMBER PATTERNS
Count with the Frog

Start here

1 A frog starts on 0 and hops 2 each time. Colour its landing places on a number strip up to 20 (or hop a counter). Write the numbers you coloured.

2 Count in 5s. Fill the gaps: 5, 10, __, 20, __, 30

3 3, 6, 9, 12, __, __. Put a counter on 3 and hop to 6, then to 9. How many steps is each hop? Write the next two numbers.

4 Write the next two numbers and the rule: 4, 7, 10, 13, __, __

Think harder

5 Fill in the missing numbers: 3, 8, __, 18, __, 28

6 Fill in the missing numbers: 40, __, 28, __, 16, 10

7 Is 30 in the pattern 4, 8, 12, 16 … ? How do you know?

What your child learns

Counting in 2s, 3s and 5s, and finding the rule of a pattern that goes up or down.

Tip for you

Hop a coin along a paper number line. Ask your child to say the rule out loud before writing the next number.

Ask

"What is the rule? Does it work for every pair?"

Go further

8 A pattern adds a number bigger than 0 each time, its 5th number is 20, and it starts at a whole number bigger than 0. How many different patterns can you make? List the first number and the rule for each.

9 Frog A starts at 3 and jumps 4 each time. Frog B starts at 5 and jumps 3 each time. Which numbers up to 40 do both frogs land on?

W1 Number Bond Inquiry

LESSON 2 • NUMBER BONDS

Start here

1 You have 10 counters. 3 are red and the rest are yellow. How many are yellow? Write 10 = 3 + __

2 A ten-frame has 6 counters in it. How many more fill it up? Write 10 = 6 + __

3 A ten-frame has 8 counters in it. How many more fill it up? Write 10 = __ + 8

4 A page has 20 stickers. 5 are stars and the rest are hearts. How many hearts? Write 20 = 5 + __

Think harder

5 5 + 4 = 6 + __

6 9 + 6 = __ + 10

7 6 + 4 = 10, so 16 + 4 = __ and 26 + 4 = __

What your child learns

Number bonds of 10 and 20, and reading = as "is the same as" on both sides.

Tip for you

Split 10 buttons into two groups in every possible way. List them in order so none is missed.

Ask

"How do you know both sides are the same?"

Go further

8 Three different numbers from 3 to 7 add to 15. What can they be? Find both ways.

9 Three different numbers from 1 to 7 add up to 10. Find every set of three.

W2

LESSON 3 • PICTURE GRAPHS

Let's Plan the Fruit Party

Start here

1 Draw a picture graph with one dot for each child: Apple 7, Banana 5, Orange 9, Grape 4. Which fruit got the most votes?

2 A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). Which fruit got the fewest votes?

3 A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). How many more children chose orange than grape? (Line the dots up side by side.)

4 A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). How many children voted altogether? (Point to each dot as you count.)

Think harder

5 1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children chose blue?

6 1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children voted altogether?

What your child learns

Reading picture graphs where one picture can stand for 1, 2 or 3 children, comparing rows and finding totals.

Tip for you

Hold a family vote (favourite fruit or snack) and draw it together, one picture per vote.

Ask

"What does one picture stand for?"

7 1 picture = 2 children. Six children chose pizza. How many pictures should we draw?

Go further

8 Votes: A 8, B 5, C 6. Then 4 more children join and all choose B. Which fruit wins now, and by how many votes?

9 1 picture = 3 children. Apple has 4 pictures, banana 2 and orange 5. Then 7 more children join and all choose banana. Which fruit wins now, and by how many votes?

W2 Pattern Inquiry

LESSON 4 • REPEATING AND GROWING PATTERNS

Start here

1 The pattern is circle, square, triangle, circle, square, triangle, circle ... What is the 10th one? (Keep writing the pattern until you get there.)

2 The pattern is star, heart, star, heart, star, heart ... What is the 9th one?

3 The pattern is red, red, blue, red, red, blue, red ... What is the 8th one?

4 The pattern is A, B, C, D, A, B, C ... What is the 11th one? (Write the letters in a row and number them.)

Think harder

5 Pattern: circle, triangle, triangle, square, circle, triangle, triangle, square ... What is the 14th shape?

6 Same pattern: how many triangles are in the first 20 shapes?

What your child learns

Finding the part of a pattern that repeats, predicting far-away places, and growing patterns with a table.

Tip for you

Make a pattern with spoons and forks. Ask what the 10th one will be before you build it.

Ask

"Which part repeats? How many things are in it?"

7 Red, blue, blue, red, blue, blue … What colour is the 20th bead?

Go further

8 Squares in a row use 4, 7, 10 … sticks. Mei says she can make a row of squares using EXACTLY 30 sticks. Can she? What about exactly 31 sticks? Explain how you know.

9 Pattern A repeats star, moon, sun. Pattern B repeats red, blue. Put them together: the 1st place is a red star, the 2nd a blue moon, the 3rd a red sun … What is the 20th place? When do we first see a red star again?

W3 Mystery Number

LESSON 5 • WORKING BACKWARDS

Start here

1 I think of a number. I add 3. I get 8. What is my number? (Hide counters under a cup and check.)

2 I think of a number. I add 6. I get 14. What is my number?

3 I think of a number. I add 5. I get 12. What is my number?

4 I think of a number. I add 9. I get 17. What is my number?

Think harder

5 Ben had some marbles. He gave 5 to Mei, then won 3. Now he has 12. How many did he start with?

6 Ben had some stickers. He gave away 6, got 4, then gave away 3. Now he has 10. How many did he have at first?

7 A number machine adds the same number each time. IN 4 gives OUT 9. What is the rule? What is OUT when IN is 12? What is IN when OUT is 20?

What your child learns

Working backwards: undoing each step, from the last step to the first.

Tip for you

Hide some counters under a cup, add a few more on top, and let your child work out how many were hidden.

Ask

"How can you check by going forwards?"

Go further

8 Zoe is 6th from the front of a queue and 6th from the back. How many children are in the queue? What if a child is 10th from the front and 10th from the back? Can you find a quick way?

9 Ali and Ben have 20 marbles together. Ali has 4 more than Ben. How many marbles does each have?

W3 LESSON 6 • MONEY Coin Combos

Start here

1 Pay exactly 15¢ using two coins from 5¢, 10¢, 20¢ and 50¢. (Use play coins.)

2 Pay exactly 40¢ using two coins from 5¢, 10¢, 20¢ and 50¢.

3 Pay exactly 55¢ using two coins from 5¢, 10¢, 20¢ and 50¢.

4 How many different ways can you make 20¢ using only 10¢ and 5¢ coins? Use play coins and write each way.

Think harder

5 Make 70¢ using exactly 4 coins from 5¢, 10¢, 20¢ and 50¢. Find every set.

6 Sam has only 10¢ and 20¢ coins. He pays exactly 90¢ using 6 coins. How many of each coin does he use?

7 What is the fewest number of coins to make 45¢?

What your child learns

Finding every way to make an amount, and paying with the fewest coins and notes, in cents and in dollars.

Tip for you

Lay real coins and notes on the table. Ask "Can you do it with fewer?" before checking the answer.

Ask

"How do you know you found them all?"

Go further

8 You have one 5¢, one 10¢ and one 20¢ coin. How many different totals can you make using one, two or all three coins? List them.

9 Which amounts in 5¢ steps (5¢, 10¢, 15¢ … 50¢) cannot be made using only 20¢ and 10¢ coins? What do you notice?

W4 LESSON 7 • 2D SHAPES
Shape Safari

Start here

1 How many sides and corners does a triangle have? Draw one and count.

2 How many sides and corners does a square have? Draw one and count.

3 Draw a pentagon (a shape with 5 straight sides). How many corners does it have?

4 Draw a hexagon (a shape with 6 straight sides). How many corners does it have?

Think harder

5 Two same-size triangles can make a square. Can they also make a bigger triangle? Draw how.

6 How many squares are in a grid 3 squares long and 2 squares high?

What your child learns

Describing shapes by sides and corners, counting every shape in a picture, and making new shapes.

Tip for you

Fold and cut a paper square. What shapes can your child make by putting the pieces together again?

Ask

"Did you count the bigger shapes too?"

7 Three squares are joined in a row to make a long rectangle. How many rectangles can you find altogether? (A square counts as a rectangle.)

Go further

8 Join 4 same-size squares, whole sides touching. How many different shapes can you make? (Turns and flips are the same shape.) How do you know you have found them all?

9 How many squares are in a grid 4 squares long and 2 squares high? What about 4 long and 3 high?

W4 LESSON 8 • LOGIC AND SORTING
Logic Lab

Start here

1 Ali is taller than Ben. Ben is taller than Mei. Who is the tallest? Who is the shortest? (Draw three people.)

2 The cat is heavier than the hen. The hen is heavier than the mouse. Who is the lightest?

3 Red, blue and green ribbons. Red is longer than blue. Green is shorter than blue. Which ribbon is the longest?

4 Ben stands in front of Mei. Mei stands in front of Sam. Who is in the middle?

Think harder

5 In a race, Sam is ahead of Tia. Tia is ahead of Raj. Mei is ahead of Sam. Who is third?

6 Ana, Bo and Cy each have a different toy: a ball, a kite or a yo-yo. Bo does not have the kite. Cy has the ball. Who has which toy?

What your child learns

Using clues to put things in order, sorting with two rules, and solving small logic grids.

Tip for you

Play "Guess my number" from 1 to 20 with yes or no questions. Which questions help most?

Ask

"Which clue tells you something for sure?"

7 Numbers 1 to 12. Which do we say when counting in 2s? Which when counting in 3s? Which in both?

Go further

8 Three boxes are labelled "Apples", "Oranges" and "Mixed". Every label is wrong. You may take one fruit from one box. Which box should you pick from, and why?

9 A number between 20 and 40 is odd, and we say it when counting in 5s. It is also one less than a number we say when counting in 4s. What is it?

W5 LESSON 9 • LENGTH
Measure It!

Start here

1 A pencil is 7 cubes long. A pen is 10 cubes long. How many cubes longer is the pen?

2 A ribbon is 12 cubes long. You cut off 5 cubes. How long is the piece left?

3 Ali's book is 9 cubes long and Mei's book is 6 cubes long. Whose is longer? By how much?

4 Put in order from shortest: 11 cubes, 7 cubes, 9 cubes, 12 cubes.

Think harder

5 A ribbon lies from the 3 cm mark to the 11 cm mark on a ruler. How long is it?

6 A red ribbon lies from the 2 cm mark to the 9 cm mark. A blue ribbon lies from the 5 cm mark to the 11 cm mark. Which ribbon is longer? By how many cm?

What your child learns

Measuring with equal units and a centimetre ruler, comparing lengths, and changing from one unit to another.

Tip for you

Measure a spoon with paper clips, then with erasers. Talk about why the two numbers are different.

Ask

"Why does a bigger unit give a smaller number?"

7 1 clip is as long as 3 cubes. A pen is 12 cubes long. How many clips long is the pen? A ruler is 5 clips long. How many cubes long is the ruler?

Go further

8 A rope is 20 cubes long. Ali cuts it into pieces that are each 4 cubes long. How many pieces does he get? How many cuts does he make? Ali says, "There is always one cut fewer than pieces." Is he right? Why?

9 A stick is 12 cubes long. Ali measures a desk by laying the stick end to end 4 times and then 5 more cubes. How long is the desk in cubes?

W5 Challenge Day

LESSON 10 • MIXED PUZZLES

Start here

1 Count in 3s: 3, 6, 9, __, __

2 10 = 4 + __ and 20 = 12 + __

3 Red 4, blue 6, green 3. How many more are blue than green?

4 I think of a number. I add 4. I get 9. What is it?

Think harder

5 A pattern starts at 4 and adds 6 each time. What is the 8th number? Is 50 in the pattern?

6 Mei thinks of a number, adds 7, takes away 2 and gets 19. Ben does the same and gets 25. How much bigger is Ben's number?

7 Ali has 5 coins worth 60¢. He has only 5¢, 10¢ and 20¢ coins. Find every possible set.

What your child learns

Choosing how to start a puzzle: draw it, list it in order, or make a table.

Tip for you

When your child is stuck, ask "Can you draw it?" before giving any other hint.

Ask

"How did you start? What would you try next time?"

Go further

8 A secret number is between 20 and 60. We say it when we count in 3s and when we count in 5s. It is also one more than a number we say when we count in 4s. What is it?

9 Triangles in a row use 3, 5, 7 … sticks. Squares in a row use 4, 7, 10 … sticks. Which numbers of sticks, up to 31, can make a row of triangles AND (with the same number of sticks) a row of squares, with none left over? What do you notice?

LOOK AFTER YOUR CHILD HAS TRIED

ALL LESSONS

Answers

Week 1 • Lesson 1: Count with the Frog

1. 2, 4, 6, 8, 10, 12, 14, 16, 18, 20.
2. 15 and 25.
3. Each hop is 3 steps. The next two numbers are 15 and 18.
4. 16, 19. Add 3.
5. 13 and 23. Add 5.
6. 34 and 22. Take away 6 (use the two neighbours 16 and 10).
7. No. The pattern goes 28, 32, so it skips 30.
8. 4 patterns: start 16 add 1; start 12 add 2; start 8 add 3; start 4 add 4. (Start 0 or a jump of 5 would need a first number of 0.)
9. 11, 23 and 35.

Week 1 • Lesson 2: Number Bond Inquiry

1. 7
2. 4
3. 2
4. 15
5. 3
6. 5
7. 20 and 30
8. 3, 5, 7 and 4, 5, 6.
9. 1, 2, 7; 1, 3, 6; 1, 4, 5; 2, 3, 5.

Week 2 • Lesson 3: Let's Plan the Fruit Party

1. Orange
2. Grape
3. 5 (9 – 4)
4. 25 (7 + 5 + 9 + 4)
5. 10
6. 24 (12 pictures, each worth 2: count in 2s)
7. 3
8. B wins with 9, which is 1 more than A.
9. Orange 15, banana 13, apple 12. Orange wins by 2.

Week 2 • Lesson 4: Pattern Inquiry

1. circle
2. star
3. red
4. C
5. triangle
6. 10. The core circle, triangle, triangle, square has 2 triangles. Count the cores in 4s: 4, 8, 12, 16, 20, so there are 5 cores. Count the triangles in 2s: 2, 4, 6, 8, 10.
7. blue
8. No for 30: the stick numbers go 4, 7, 10, 13, 16, 19, 22, 25, 28, 31 and skip 30 (each number is 1 more than a number we say counting in 3s, and 29 is not one). Yes for 31: that is 10 squares.
9. The 20th place is a blue moon (20 is 2 after 18, so moon; 20 is even, so blue). A red star comes back at the 7th place, then the 13th: every 6 places (3 and 2 meet at 6).

Week 3 • Lesson 5: Mystery Number

1. 5
2. 8
3. 7
4. 8
5. 14 (12 – 3 + 5)
6. 15 (work backwards: 10 + 3 = 13, 13 – 4 = 9, 9 + 6 = 15; check 15 – 6 + 4 – 3 = 10)
7. Add 5. OUT is 17 when IN is 12. IN is 15 when OUT is 20.
8. 11 and 19. Zoe is counted twice if we add 6 + 6, so take away 1. Quick way: double the place number and take away 1 (6 + 6 – 1 = 11; 10 + 10 – 1 = 19).
9. Ben 8 and Ali 12. (Take away the extra 4: 16 left, share: 8 and 8, give Ali the 4 back: 12.)

Week 3 • Lesson 6: Coin Combos

1. 5¢ + 10¢
2. 20¢ + 20¢
3. 5¢ + 50¢
4. 3 ways: 10 + 10; 10 + 5 + 5; 5 + 5 + 5 + 5.
5. 2 sets: 50¢ + 10¢ + 5¢ + 5¢, and 20¢ + 20¢ + 20¢ + 10¢.
6. Three 10¢ and three 20¢ (30 + 60 = 90).
7. 3 coins (20¢ + 20¢ + 5¢)
8. 7 totals: 5, 10, 15, 20, 25, 30, 35.
9. 5¢, 15¢, 25¢, 35¢ and 45¢. (Every amount that does not end in 0.)

Week 4 • Lesson 7: Shape Safari

1. 3 sides and 3 corners
2. 4 sides and 4 corners
3. 5 corners
4. 6 corners
5. Yes. Join them along a short side (the two equal short sides touching) to make one bigger triangle. Joining along the long side makes a square instead.
6. 8 (6 small squares + 2 squares made of 4 small ones)
7. 6 (3 single squares, 2 pairs and 1 whole row).
8. 5 shapes: a line of 4, a 2 by 2 square, an L, a T and a zigzag. Build them in order: start with a line of 4, then move one square at a time and check each new one is not a turned or flipped copy.
9. 11 (8 small + 3 of size 2 by 2) and 20 (12 small + 6 of size 2 by 2 + 2 of size 3 by 3).

Week 4 • Lesson 8: Logic Lab

1. Ali is tallest; Mei is shortest.
2. The mouse
3. Red
4. Mei
5. Tia (order: Mei, Sam, Tia, Raj).
6. Cy: ball. Bo: yo-yo. Ana: kite.
7. Counting in 2s: 2, 4, 6, 8, 10, 12. Counting in 3s: 3, 6, 9, 12. Both: 6 and 12.
8. Pick from the box labelled "Mixed". Its label is wrong, so it holds only one kind of fruit. If you take out an apple, that box is Apples. The box labelled "Oranges" cannot be oranges, and apples are taken, so it is Mixed. The box labelled "Apples" must be Oranges.
9. 35. The odd ones are 25 and 35. 25 + 1 = 26: not said counting in 4s. 35 + 1 = 36: yes.

Week 5 • Lesson 9: Measure It!

1. 3
2. 7 cubes
3. Ali's, by 3 cubes.
4. 7, 9, 11, 12
5. 8 cm (count the spaces from 3 to 11, or 11 − 3)
6. Red is longer: red 7 cm, blue 6 cm, so red is 1 cm longer.
7. 4 clips. 15 cubes.
8. 5 pieces and 4 cuts. Yes: the last piece needs no cut, so cuts are always 1 fewer than pieces.
9. 53 cubes (12 + 12 + 12 + 12 + 5).

Week 5 • Lesson 10: Challenge Day

1. 12, 15
2. 6 and 8
3. 3
4. 5
5. 46 (4, 10, 16, 22, 28, 34, 40, 46). No: the pattern goes 46, 52, so it skips 50.
6. 6 bigger. Work backwards: Mei 19 + 2 = 21, 21 − 7 = 14; Ben 25 + 2 = 27, 27 − 7 = 20; 20 − 14 = 6. (Shortcut: both did the same steps, so the answers differ by the same amount: 25 − 19 = 6.)
7. Two sets: 20 + 10 + 10 + 10 + 10, and 20 + 20 + 10 + 5 + 5. Make a table by number of 20¢ coins: one 20¢ needs 40¢ from 4 coins (four 10¢); two 20¢ need 20¢ from 3 coins (10 + 5 + 5); three 20¢ need 0¢ from 2 coins (impossible); no 20¢: 5 coins make at most 50¢.
8. 45. Numbers in both the 3s and the 5s counts: 30, 45, 60. 30 − 1 = 29 and 60 − 1 = 59 are not said counting in 4s; 45 − 1 = 44 is.
9. 7, 13, 19, 25 and 31. They go up by 6 (the 2 and the 3 from the two patterns meet every 6). For example 7 sticks make 3 triangles or 2 squares.